

Afsal V N

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Kochi, Kerala

EDUCATION

College of Engineering Trivandrum (CET)

BTech in Electrical and Electronics Engineering | CGPA: 8.27/10

Trivandrum, Kerala

2023 – 2027

St. Anns Public School

Grade 12 (CBSE) | 94%

Kochi, Kerala

2023

EXPERIENCE

BSSR Lab, Amrita Vishwa Vidyapeetham

Research Intern

Coimbatore, Tamil Nadu

May 2026 – Jun 2026

- Designed and integrated EEG signal-acquisition electronics with a Raspberry Pi to drive real-time prosthetic robotic actuators in a Brain-Computer Interface system.
- Built the embedded signal pipeline – acquisition, preprocessing, and on-device classification – validating output through hardware-in-loop testing and system-level verification.

C2S Lab, CET

Technical Intern

Trivandrum, Kerala

2025

- Designed and synthesized digital logic circuits in Verilog HDL on FPGA, covering RTL electronics design through synthesis and implementation.
- Performed functional verification, simulation, and hardware testing to validate circuit behaviour in the FPGA prototyping flow.

PROJECTS

UART Implementation on FPGA | Verilog HDL, Xilinx Basys 3 FPGA

- Designed a parameterized UART transceiver with FSM-based TX/RX logic and a custom baud rate generator.
- Verified data integrity through simulation and on-board hardware loopback testing.

InnoPlay – Gesture Gaming Controller | ESP32-S3, Embedded C, Edge Impulse

- Built a gesture-based gaming controller for people with limited mobility using XIAO ESP32-S3 Sense and Grove AI Vision V2.
- Trained and deployed machine learning models in Edge Impulse for real-time, on-device inference.

TECHNICAL SKILLS

- Hardware & Embedded:** ESP32-S3, Raspberry Pi, Arduino, Microcontrollers, FPGA Prototyping
- Programming & HDL:** Embedded C, C, Verilog HDL, Python
- EDA & Simulation Tools:** Proteus, MATLAB, Simulink, Electric VLSI, Edge Impulse, Git
- Concepts:** Electrical Systems, Digital Electronics Design, Embedded Systems, Hardware Verification & Testing, Signal Processing

LEADERSHIP & ACTIVITIES

IEEE Student Branch, CET

Chairperson

Trivandrum, Kerala

2025 – Present

- Lead a team of 80+ members and have organized 60+ technical events and workshops.

RoboCET

General Secretary

Trivandrum, Kerala

2024 – Present

- Lead a technical team of 40+ students and organized 12+ hands-on hardware and robotics workshops.

ACHIEVEMENTS

- TinyML Hackathon:** Won the Jury Award for ‘InnoPlay’, a gesture-controlled gaming controller.
- Outstanding Student Volunteer Award** – IEEE Student Branch CET, 2025.

CERTIFICATIONS

- ISWDP Level 1** – India Semiconductor Workforce Development Program (IISc Bangalore, Samsung Semiconductor, Synopsys).
- VLSI Chip Design and Simulation** – L&T EduTech.
- Hardware Modeling Using Verilog** – NPTEL.